

1. Introduction - The Basics

In this class we cover the basics and provide a few expressions for electrostatic force, electric field, etc.

Let us consider two points A and B , containing point charges Q_A and Q_B in vacuum, on positions $\vec{r}_A$ and $\vec{r}_B$ respectively w.r.t the origin, O . We will derive expressions for electrostatic force, electric field, and electric potential for an arbitrary point C in the same vacuum at position $\vec{r}_C$ w.r.t the same origin, O . Figure 1 illustrates the points A , B , and C along with their position vectors.

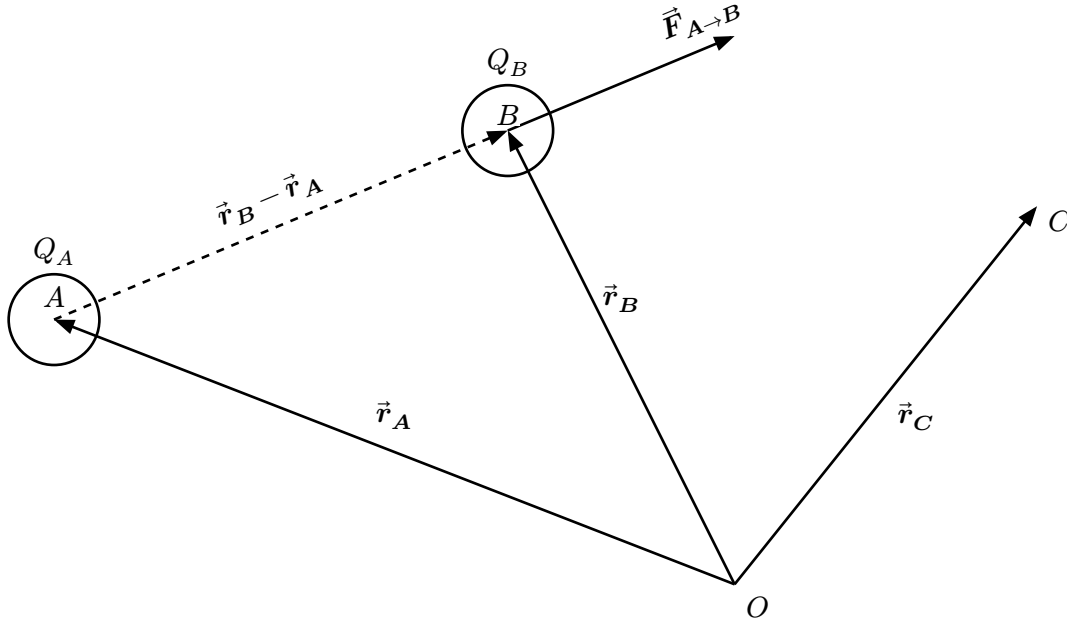


Figure 1: The figure illustrates points A , B , C , and the origin, O , in vacuum. $\vec{r}_A$, $\vec{r}_B$, and $\vec{r}_C$ represents the position vectors of points A , B , and C respectively w.r.t O . Q_A and Q_B represents the point charges placed on A and B respectively. $\vec{F}_{A \rightarrow B}$ represents the electrostatic force on the point charge placed at B due to the point charge placed at A .

The electrostatic force acting on the point charge placed at B due to the presence of the point charge placed at A will be represented by $\vec{F}_{A \rightarrow B}$. Using Coulumb's Law the expression for this electrostatic force can be written as

$$\vec{F}_{A \rightarrow B} = \frac{1}{4\pi\epsilon_0} Q_B Q_A \frac{\vec{r}_B - \vec{r}_A}{|\vec{r}_B - \vec{r}_A|^3}$$

where ϵ_0 represents the permittivity constant of vacuum. Similarly, the electrostatic force acting on the point charge placed at A due to the presence of the point charge placed at B will be represented by $\vec{F}_{B \rightarrow A}$ and the expression for this electrostatic force can be written as

$$\vec{F}_{B \rightarrow A} = -\vec{F}_{A \rightarrow B} = \frac{1}{4\pi\epsilon_0} Q_A Q_B \frac{\vec{r}_A - \vec{r}_B}{|\vec{r}_A - \vec{r}_B|^3}$$

We can take this even further and calculate the electrostatic force acting on an arbitrary charge Q_C placed on point C , whose position is represented by $\vec{r}_C$, due to the presence of the charges Q_A and Q_B at points A and B respectively.

$$\begin{aligned}\vec{F}_{(A,B) \rightarrow C} &= \vec{F}_{A \rightarrow C} + \vec{F}_{B \rightarrow C} \\ &= \frac{1}{4\pi\epsilon_0} Q_C Q_A \frac{\vec{r}_C - \vec{r}_A}{|\vec{r}_C - \vec{r}_A|^3} + \frac{1}{4\pi\epsilon_0} Q_C Q_B \frac{\vec{r}_C - \vec{r}_B}{|\vec{r}_C - \vec{r}_B|^3}\end{aligned}$$

This can be extended to any number of charges placed in the environment. For example, the electrostatic force acting on the charge Q_C placed at the point C due to a collection of N charges $\{Q_i\}_{i=1}^N$ placed at points $\{P_i\}_{i=1}^N$ whose positions are given by the vectors $\{\vec{r}_i\}_{i=1}^N$ respectively can be calculated as follows

$$\vec{F}_{\{P_i\}_{i=1}^N \rightarrow C} = \sum_{i=1}^N \frac{1}{4\pi\epsilon_0} Q_C Q_i \frac{\vec{r}_C - \vec{r}_{P_i}}{|\vec{r}_C - \vec{r}_{P_i}|^3}$$

Let us consider another example where we calculate the electrostatic force on an arbitrary charge Q_C placed at point C due to an arbitrary charged body P with a charge density function ρ . This electrostatic force, $\vec{F}_{P \rightarrow C}$ can be calculated as follows

$$\vec{F}_{P \rightarrow C} = \int_V \frac{1}{4\pi\epsilon_0} Q_C \rho(\vec{r}') \frac{\vec{r}_C - \vec{r}'}{|\vec{r}_C - \vec{r}'|^3} d^3\vec{r}'$$

where $\rho(\vec{r}')$ represents the charge density of the infinitesimally small volume $d^3\vec{r}'$ located at position $\vec{r}'$.

We have seen that the electrostatic force acting on a charge (say q) placed at an arbitrary point P depends heavily on the charges already present in the environment. In fact, the magnitude of the charge q only acts as a scaling factor to the force acting at position P due to the charges already present in the environment. We introduce a concept called the *Electric Field* at the point P due to some charge/charges already present in an environment to explore this observation. At this moment we can define the *Electric Field* at a point whose position is represented by $\vec{r}$ as the electrostatic force acting on an unit charge placed at that point. Mathematically we can write the electric field at a point in position $\vec{r}$ due to a point charge Q_X at point X is as follows

$$\vec{E}_X(\vec{r}) = \lim_{q \rightarrow 0} \frac{\vec{F}(\vec{r})}{q}$$

where $\vec{F}(\vec{r})$ represents the electrostatic force on the charge q at position $\vec{r}$.

Now for a simple point charge like Q_A and Q_B the electrostatic force acting on them is simply the electric field scaled with their individual charge. For example, the electrostatic force on charge Q_A due to charge Q_B , $\vec{F}_{B \rightarrow A}$ is

$$\vec{F}_{B \rightarrow A} = Q_A \vec{E}_B(\vec{r}_A)$$

where

$$\vec{E}_B(\vec{r}_A) = \frac{1}{4\pi\epsilon_0} Q_B \frac{\vec{r}_A - \vec{r}_B}{|\vec{r}_A - \vec{r}_B|^3}$$

and similarly we have $\vec{F}_{A \rightarrow B}$ as

$$\vec{F}_{A \rightarrow B} = Q_B \vec{E}_A(\vec{r}_B)$$

where

$$\vec{E}_A(\vec{r}_B) = \frac{1}{4\pi\epsilon_0} Q_A \frac{\vec{r}_B - \vec{r}_A}{|\vec{r}_B - \vec{r}_A|^3}$$

We can take this further for more complicated distributions of charges. For example, the electric field generated by charges Q_A and Q_B at the position $\vec{r}_C$ can be written as

$$\begin{aligned} \vec{E}_{\{A,B\}}(\vec{r}_C) &= \vec{E}_A(\vec{r}_C) + \vec{E}_B(\vec{r}_C) \\ &= \frac{1}{4\pi\epsilon_0} \left\{ Q_A \frac{\vec{r}_C - \vec{r}_A}{|\vec{r}_C - \vec{r}_A|^3} + Q_B \frac{\vec{r}_C - \vec{r}_B}{|\vec{r}_C - \vec{r}_B|^3} \right\} \end{aligned}$$

As Q_C is a point charge we can calculate the electrostatic force on it as

$$\begin{aligned} \vec{F}_{\{A,B\} \rightarrow C} &= Q_C \vec{E}_{\{A,B\}}(\vec{r}_C) \\ &= \frac{1}{4\pi\epsilon_0} Q_C \left\{ Q_A \frac{\vec{r}_C - \vec{r}_A}{|\vec{r}_C - \vec{r}_A|^3} + Q_B \frac{\vec{r}_C - \vec{r}_B}{|\vec{r}_C - \vec{r}_B|^3} \right\} \end{aligned}$$

Similarly, we can also calculate the electric field at position $\vec{r}$ generated by a collection of N point charges $\{Q_i\}_{i=1}^N$ each positioned at $\vec{r}_i$ with $i = 1, 2, \dots, N$ as

$$\begin{aligned} \vec{E}_{\{Q_i\}_{i=1}^N}(\vec{r}) &= \sum_{i=1}^N \vec{E}_{Q_i}(\vec{r}) \\ &= \frac{1}{4\pi\epsilon_0} \sum_{i=1}^N Q_i \frac{\vec{r} - \vec{r}_i}{|\vec{r} - \vec{r}_i|^3} \end{aligned}$$

Now, the electric field on a position $\vec{r}$ generated by a continuous distribution of charge, represented by a charge distribution function ρ , can be written as

$$\vec{E}_{\rho(\Omega)}(\vec{r}) = \frac{1}{4\pi\epsilon_0} \int_{\Omega} \rho(\vec{r}') \frac{\vec{r} - \vec{r}'}{|\vec{r} - \vec{r}'|^3} d^3\vec{r}'$$